



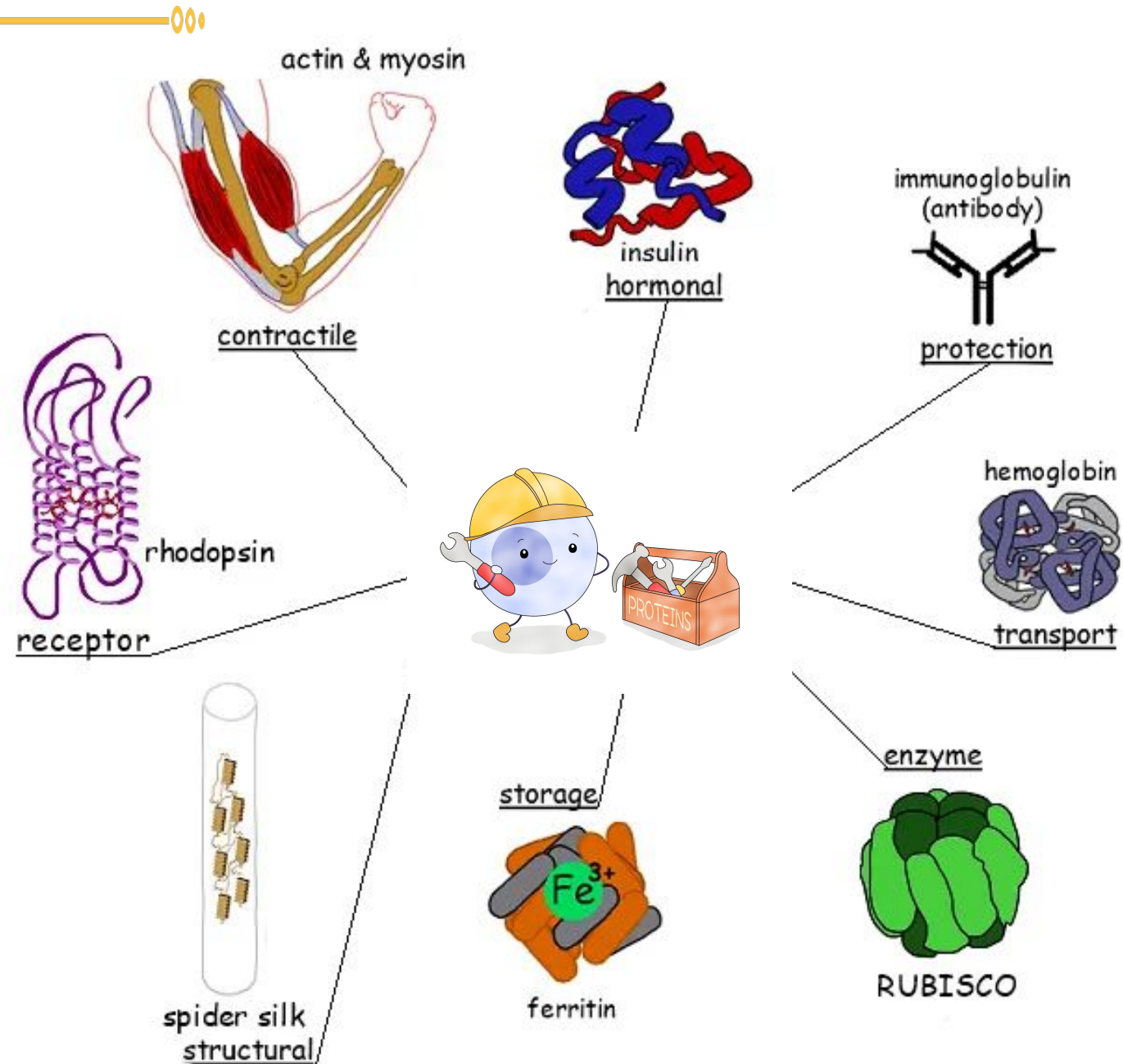
Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-9: 2 Classes

*Please listen Carefully
&*

Please keep your phone(s) on silent mode

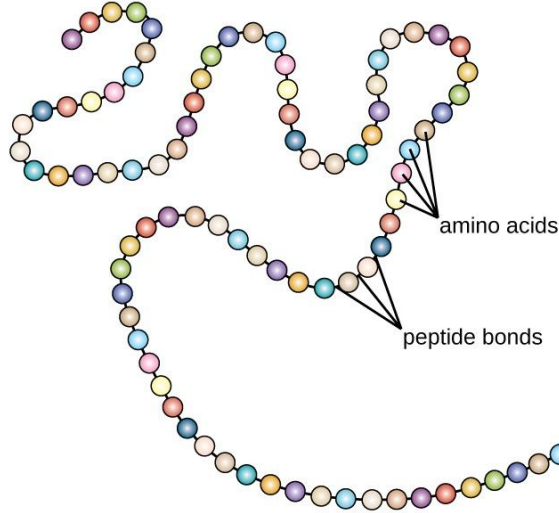
Protein as the workforce



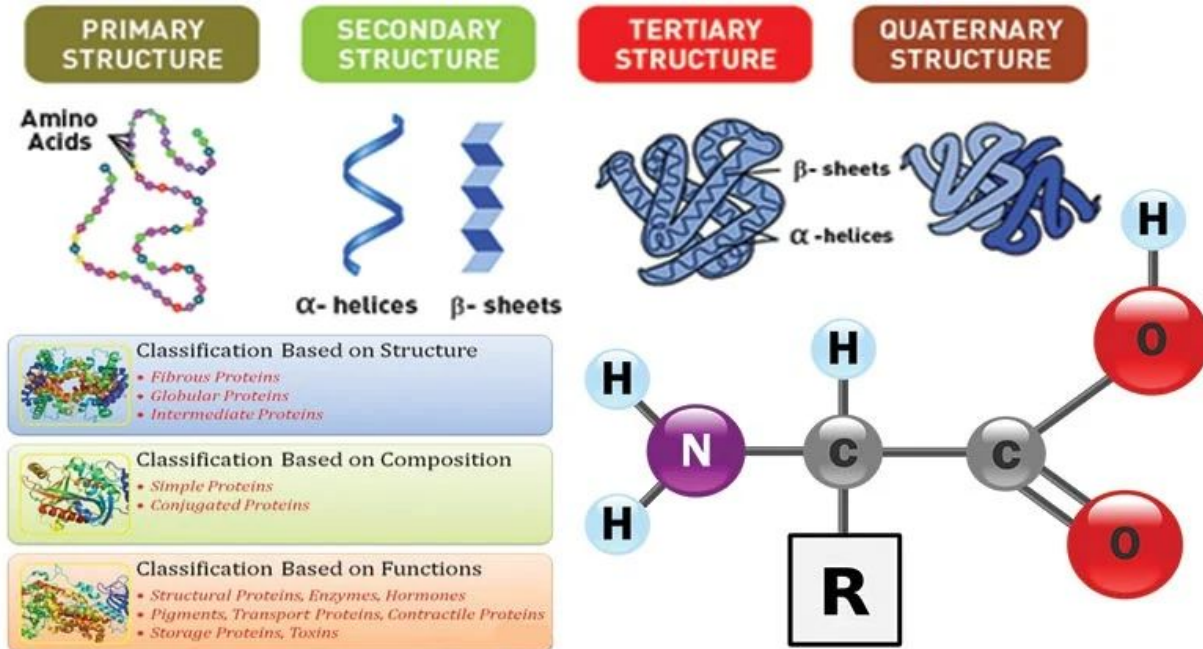
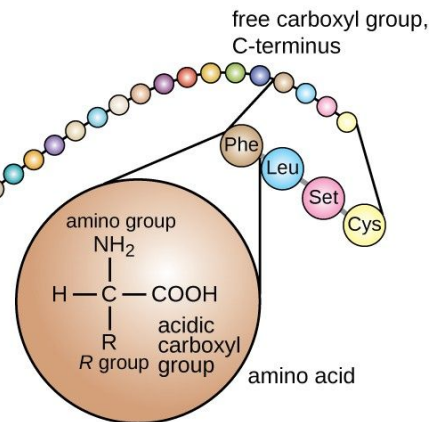
Properties of proteins

- Found in all living organisms
- Involved in processes such as digestion of food, cell structure, catalysis, movement, energy and manipulation.
- Complex molecules; Polymers of amino acids – polypeptides
- Three-dimensional structure
- Composition
- Way how proteins fold

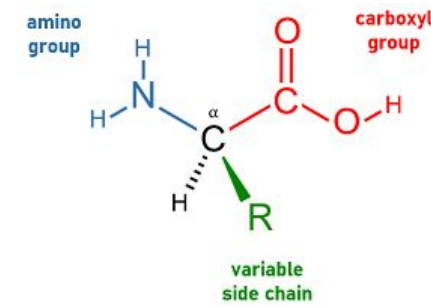
free amino group,
N-terminus



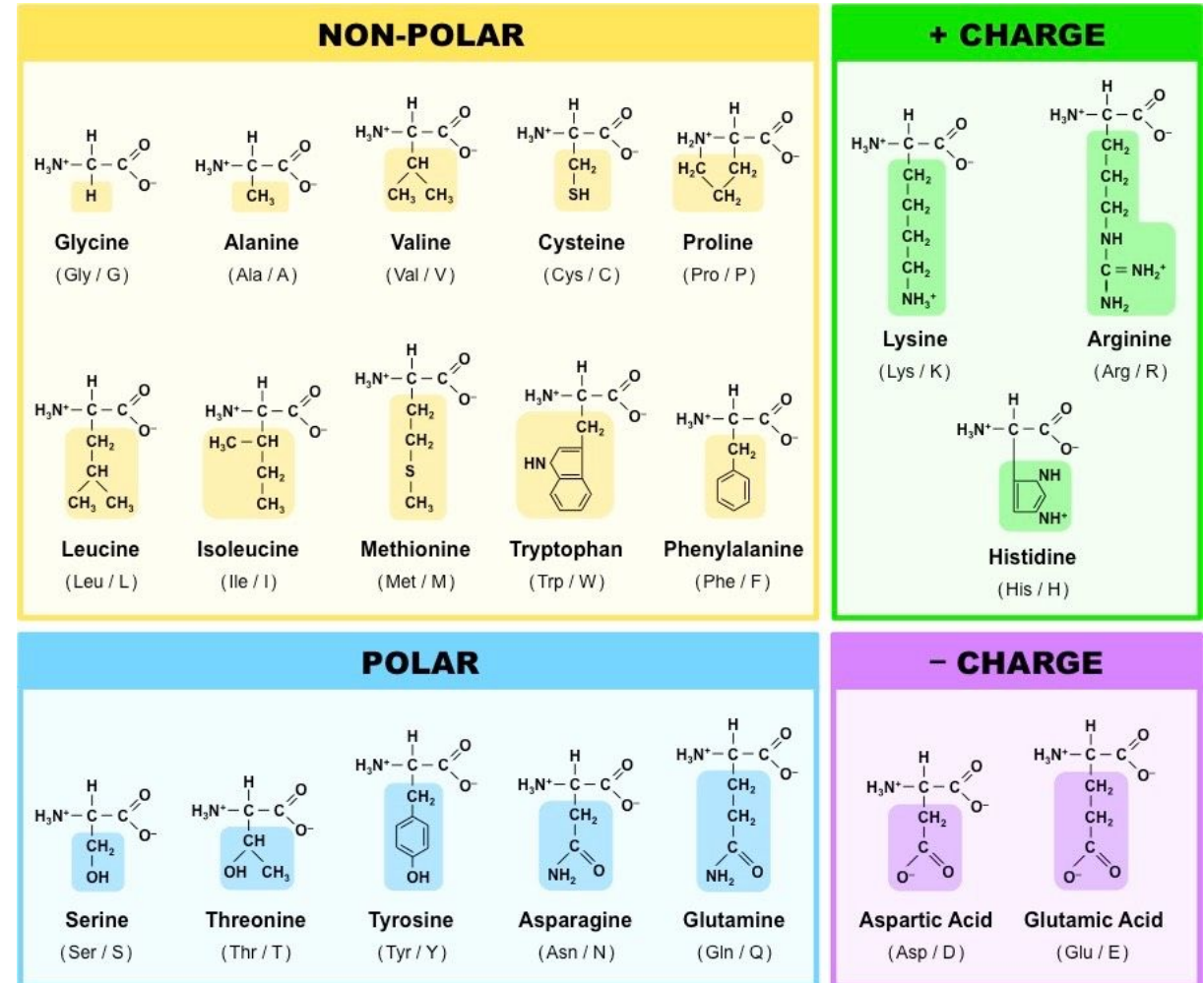
The primary protein structure is the chain of amino acids that makes up the protein.



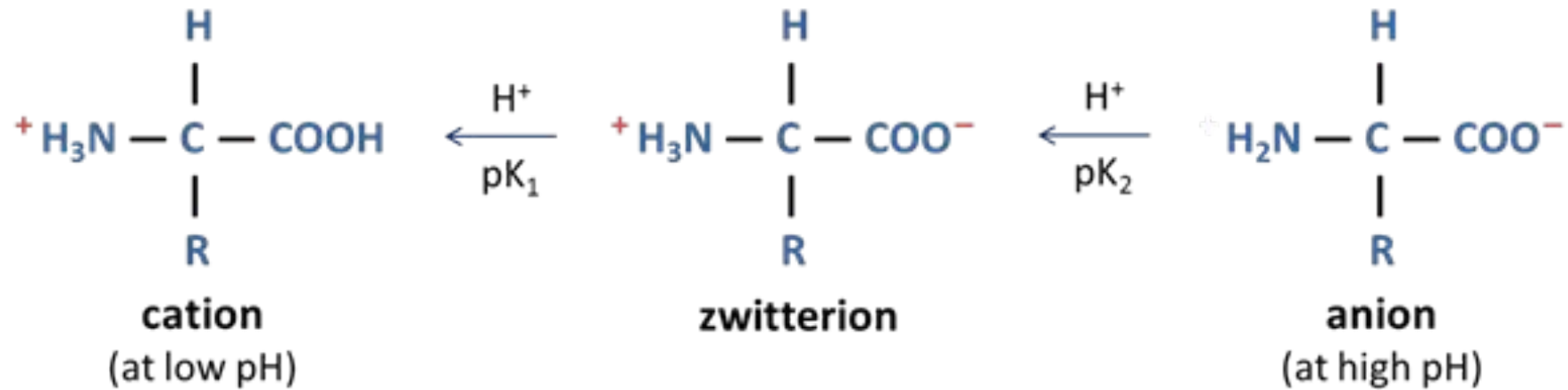
Amino Acids



- Fundamental **building blocks of proteins**.
- Have a common structure consisting of an **amino group** ($-\text{NH}_2$), a **carboxyl group** ($-\text{COOH}$), a **hydrogen atom**, and a **unique side chain** (R group).
- There are **20 standard amino acids** used in protein synthesis, each characterized by a different R group.
- Amino acids are categorized as **essential** (must be obtained from the diet) and **non-essential** (can be synthesized by the body).
- Amino acids have various roles in the body, including serving as building blocks for proteins, participating in metabolic pathways, and acting as neurotransmitters.

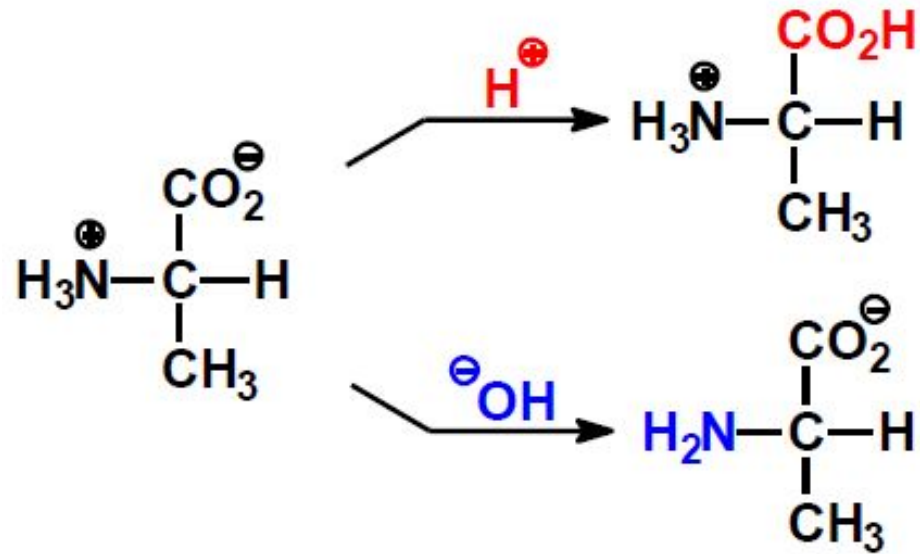


Amino Acids as zwitterions



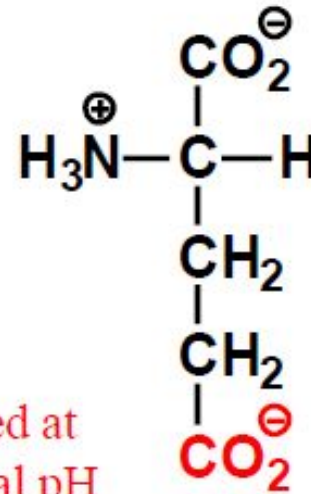
- Amino acids exhibit zwitterionic properties, meaning they **can exist** as both a **positively charged ion** (cation) and a **negatively charged ion** (anion) within the same molecule.
- This zwitterionic nature is **critical to the behavior of amino acids** in biological systems. It affects their **solubility in water**, their **ability to form hydrogen bonds with other molecules**, and their **participation in biochemical reactions (Isoelectric point)**.
- When amino acids are incorporated into proteins, their zwitterionic properties play a

Amino Acids can react as either acids or bases



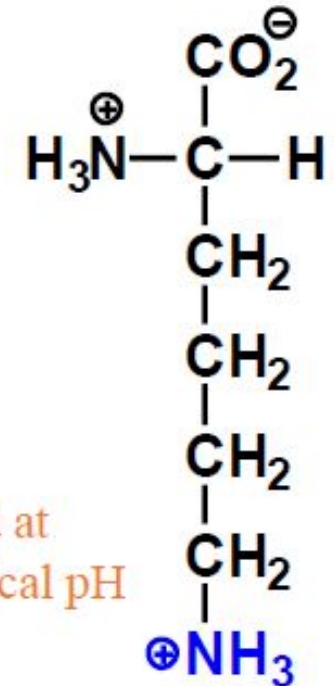
Amino acids can also have side chains containing acidic or basic groups

Deprotonated at physiological pH



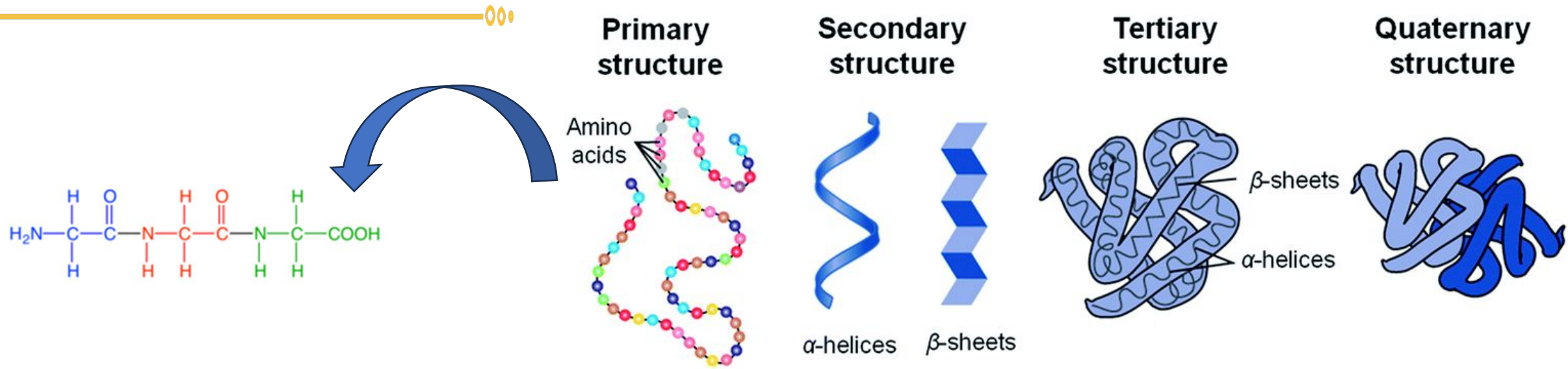
glutamic acid – acidic R

Protonated at physiological pH



Lysine –basic R

Protein structure



1. Primary Structure: Linear sequence of amino acids linked together by peptide bonds. This sequence is encoded by the organism's DNA and is unique to each protein.

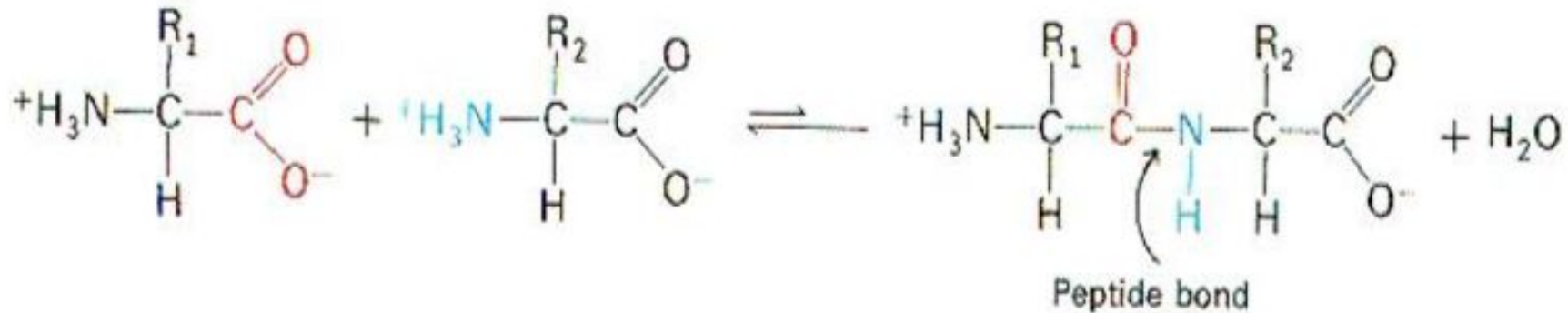
2. Secondary Structure: Proteins often adopt secondary structures, such as alpha helices and beta sheets, **due to hydrogen bonding between nearby amino acids**. These structures result from folding patterns within the linear sequence.

3. Tertiary Structure: The three-dimensional arrangement of a protein's secondary structures and the interactions between distant amino acids define its tertiary structure. These interactions include **hydrogen bonds, disulfide bridges, hydrophobic interactions, and electrostatic forces**.

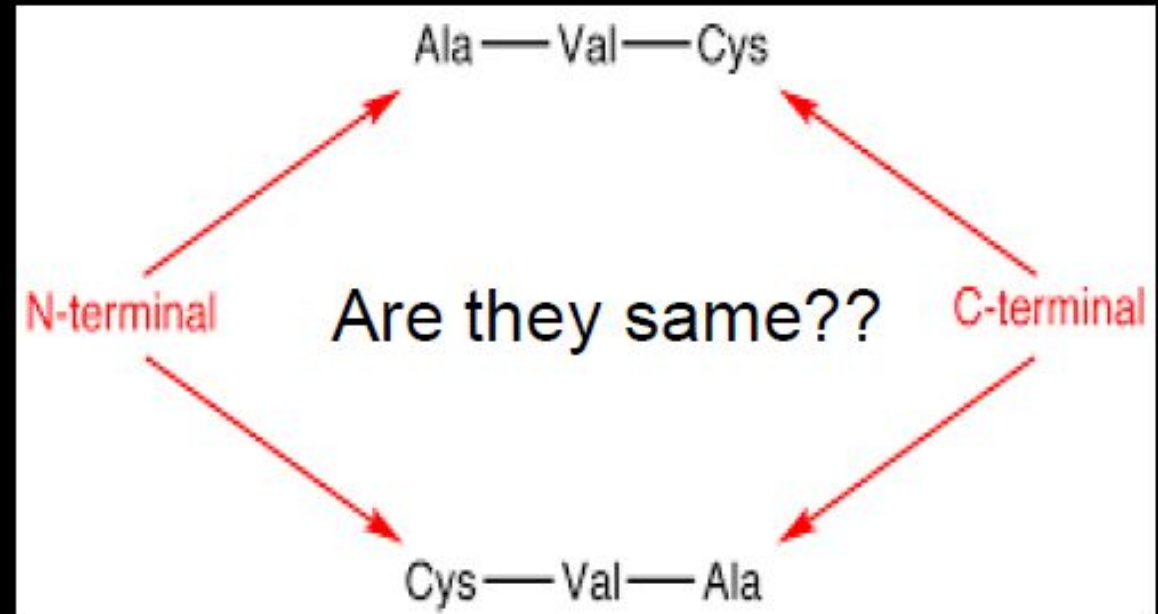
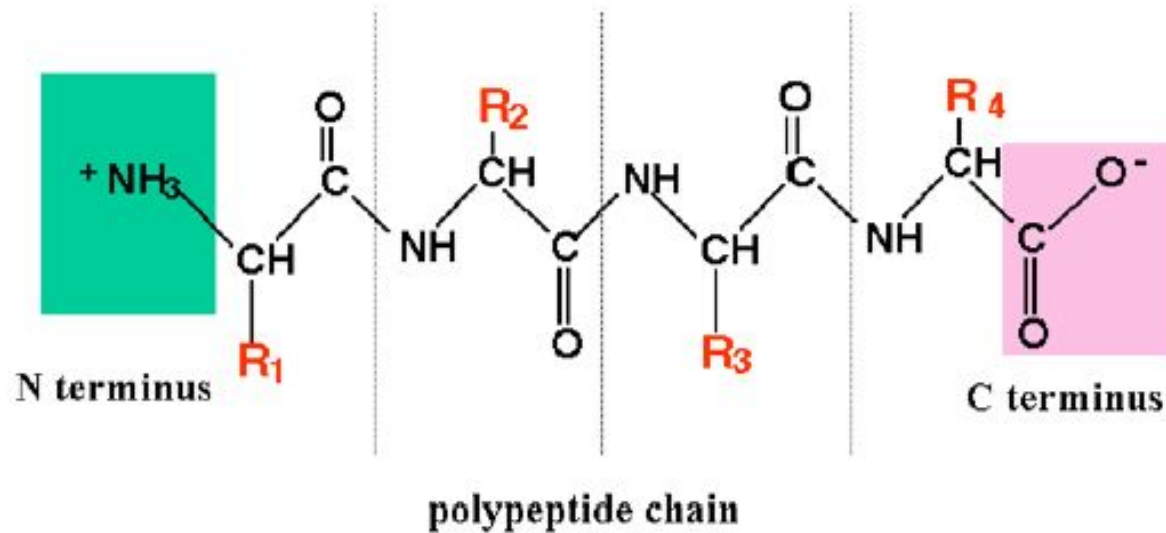
4. Quaternary Structure: Some proteins consist of **multiple polypeptide chains**, known as subunits, **which assemble**

Joining the amino acids

00



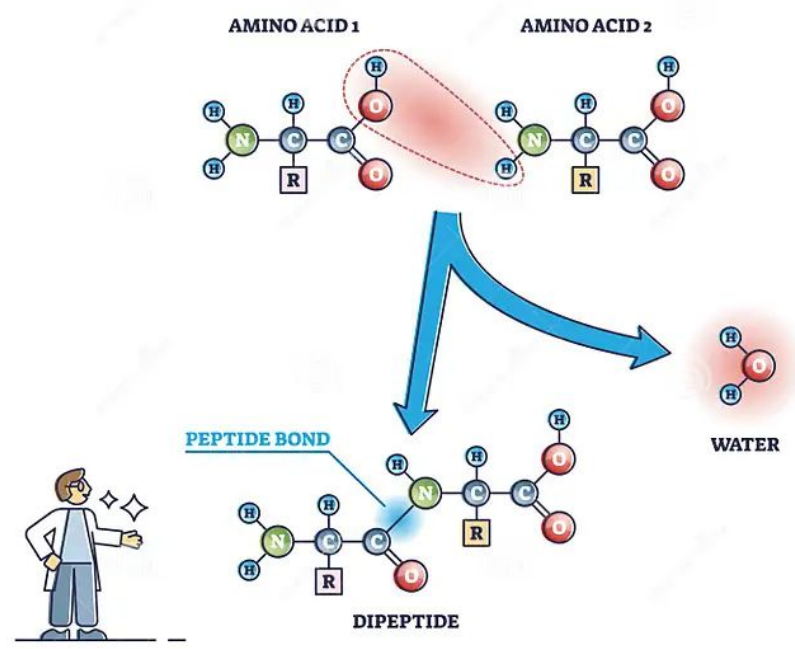
Peptide = chain of amino acids



Peptide Bond, 2^o structure and RC plot

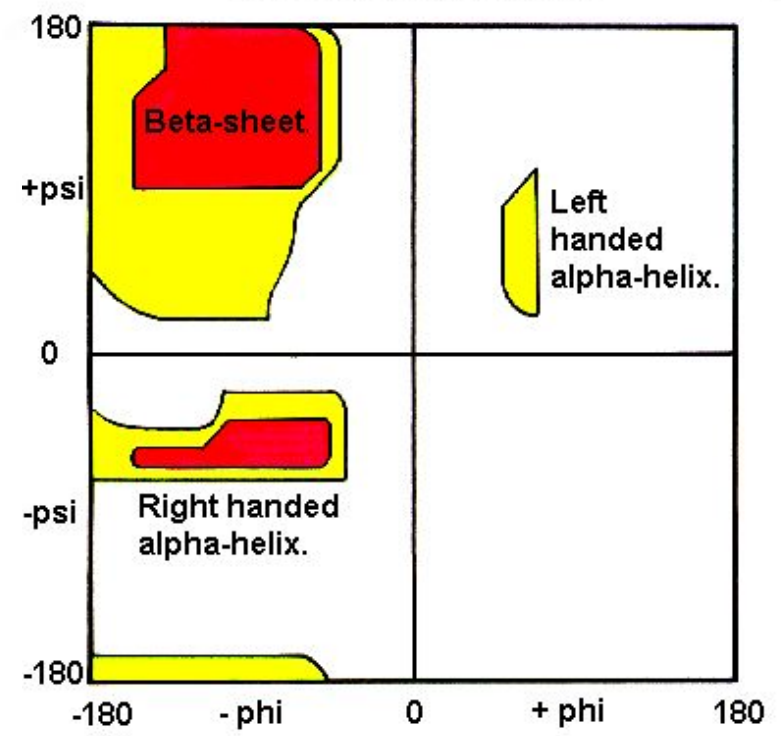
000

PEPTIDE BOND



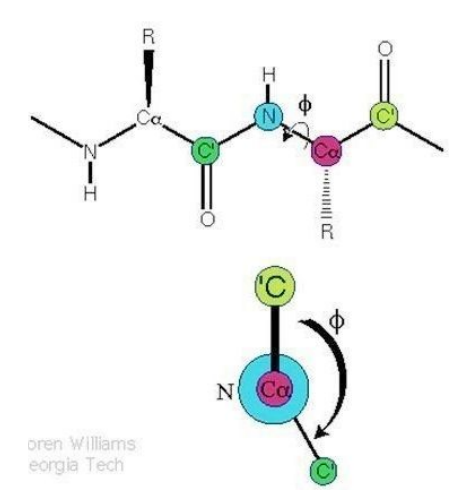
Red region- no steric clashes
Yellow region- limited clashes
White region- no allowed structures

The Ramachandran Plot.



Torsion Angles Phi and Psi

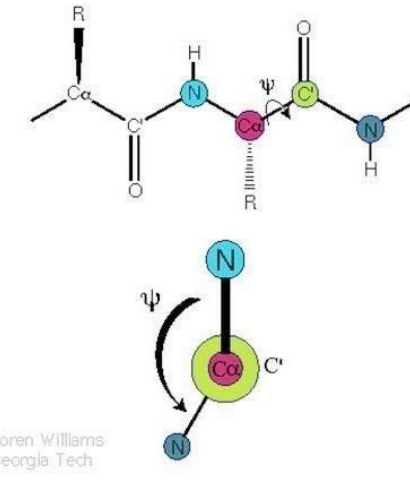
Torsion Angle Phi (ϕ)



Loren Williams
Georgia Tech

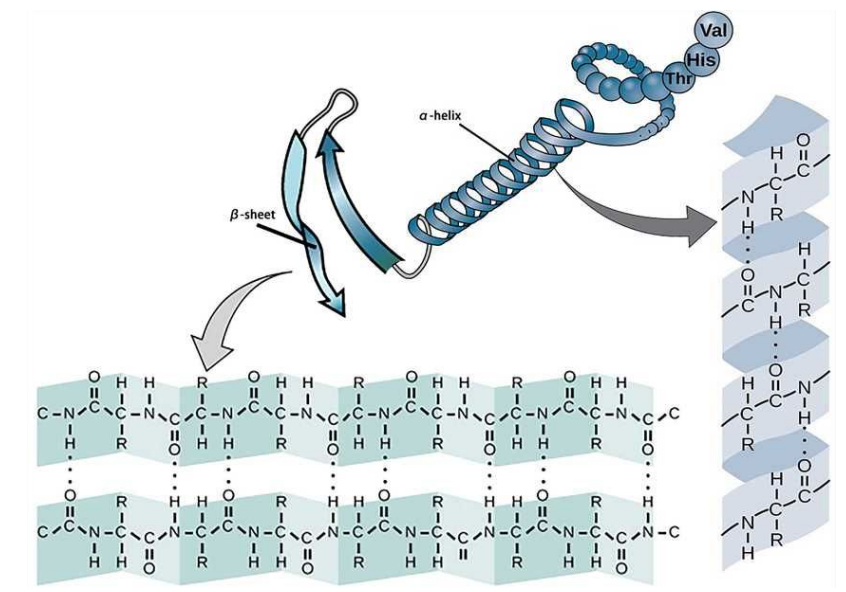
This torsion angle ϕ is right handed and is therefore positive (+130°)

Torsion Angle Psi (ψ)

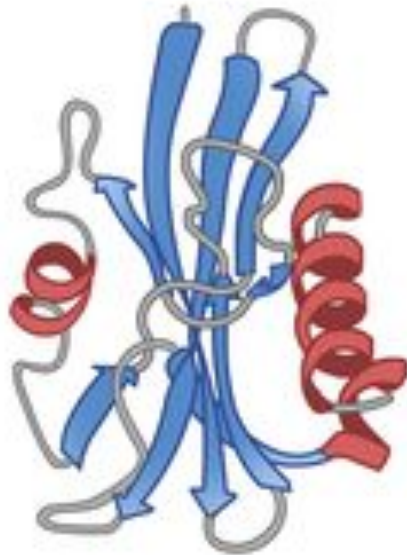


Loren Williams
Georgia Tech

This torsion angle ψ is left-handed and is therefore negative (-130°)



Tertiary and Quaternary structure



Tertiary Structure
Three-dimensional structure

- Represents its **three-dimensional arrangement in space** and is a critical determinant of its function.
- **Arises from the intricate folding** and **interactions of secondary structural elements** and the various side chains of amino acids.
- Determines **how the protein interacts with other molecules**, allowing it to carry out its biological role effectively.



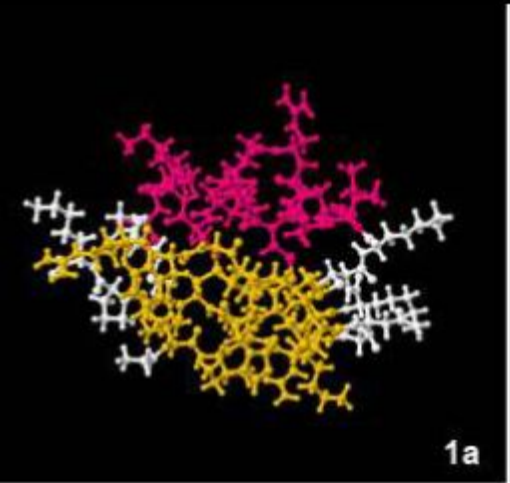
Quaternary Structure
complex of protein molecules

- Represents a **higher-order organization of multiple polypeptide chains** to form a functional, biologically active protein complex.
- Allows for the **cooperative functioning of subunits** within a protein complex.
- It often **enhances the efficiency and versatility** of the protein's activity.

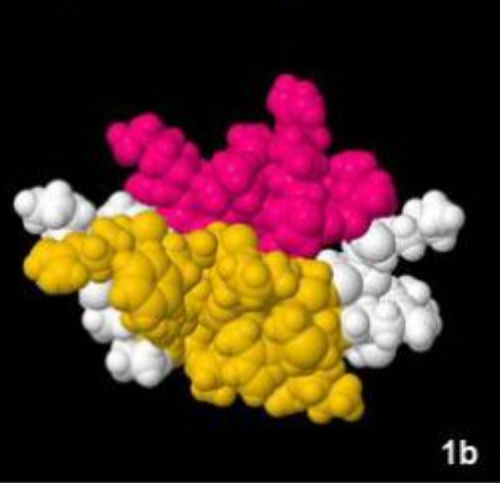
3D structure: Way to represent proteins



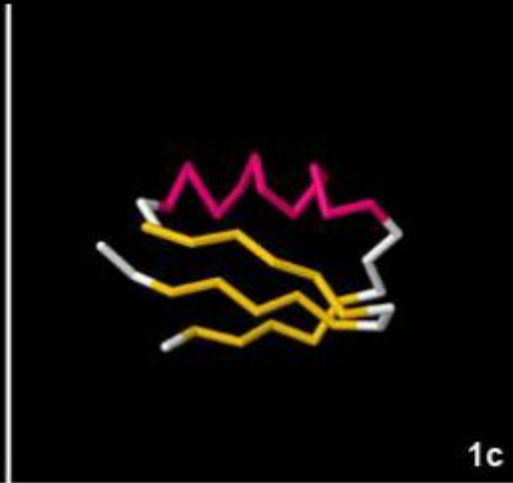
Ball and stick model



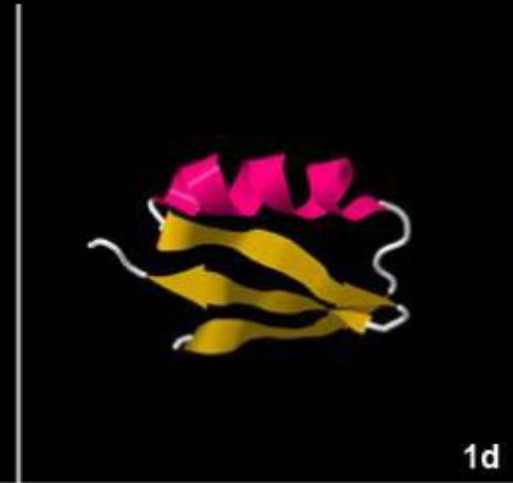
Space filling model



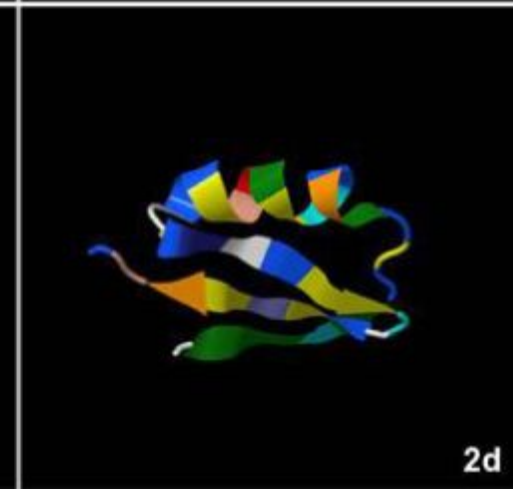
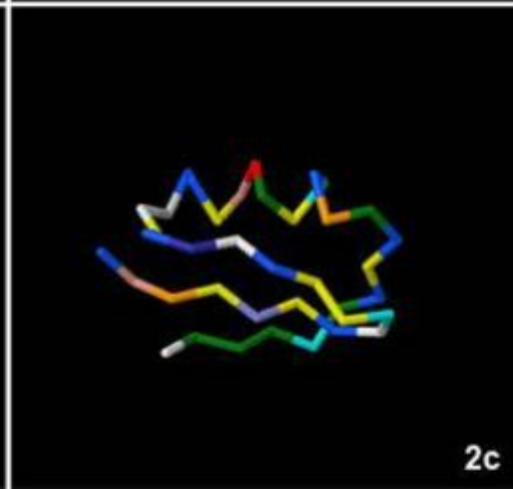
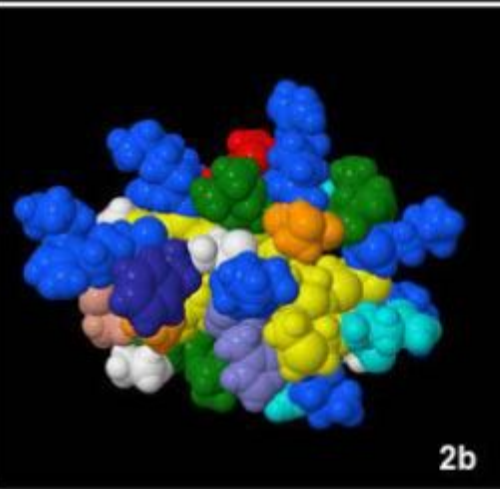
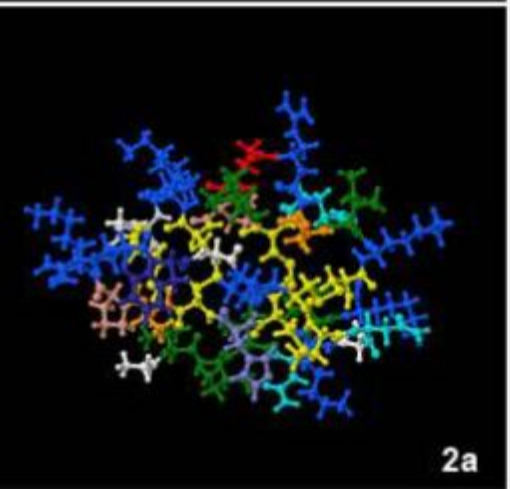
Backbone model



Ribbon (or cartoon) model

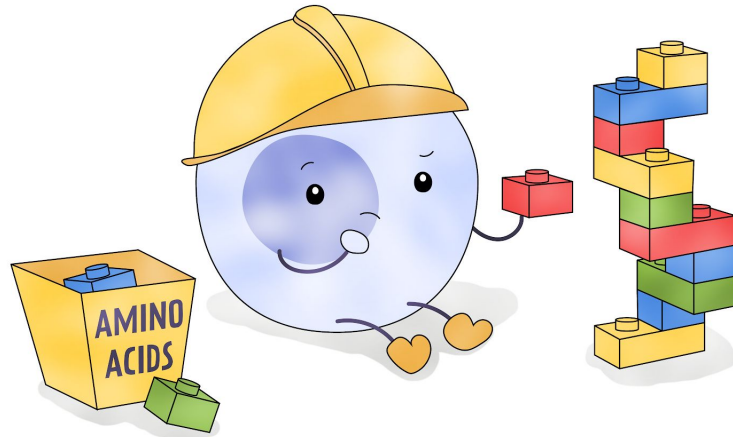


Secondary Structure

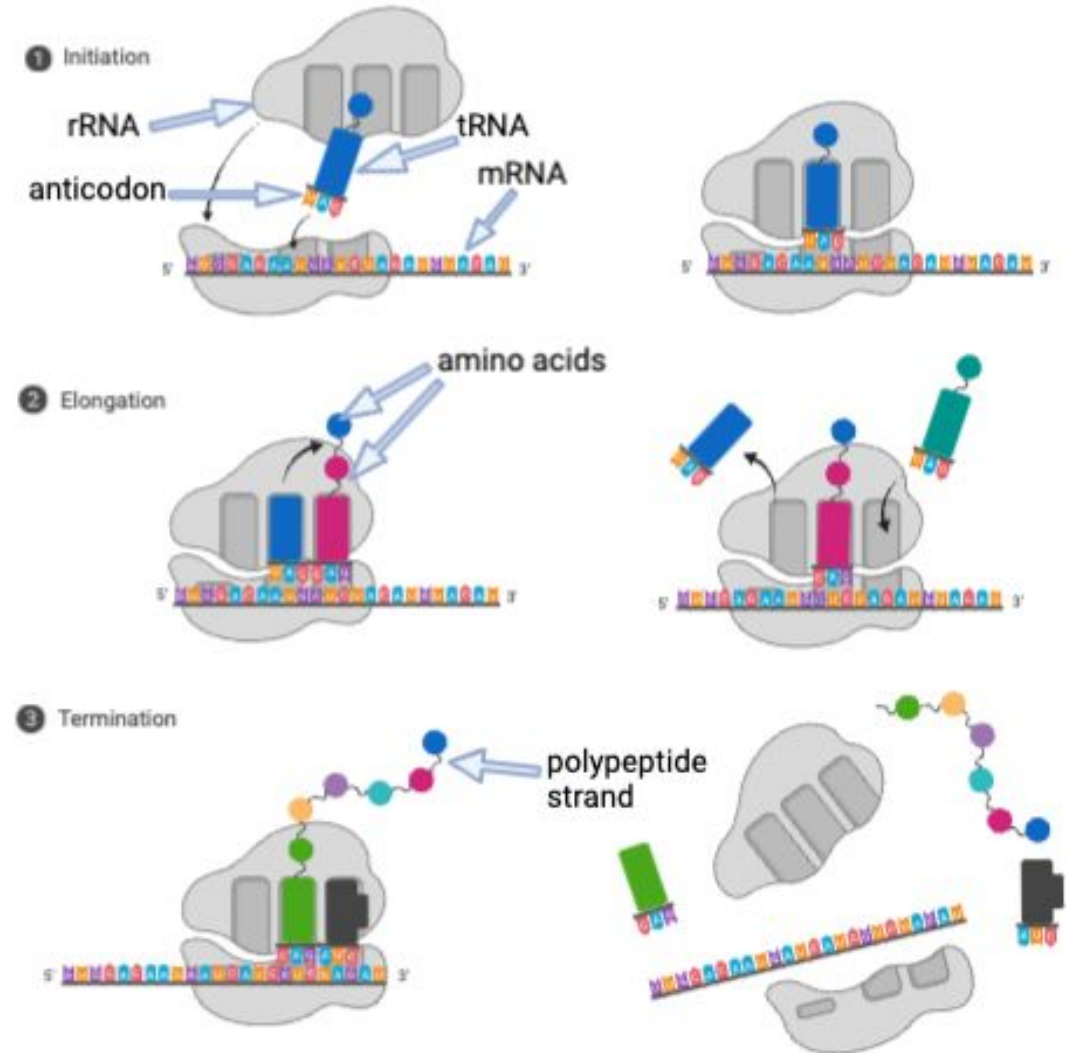


All amino acids

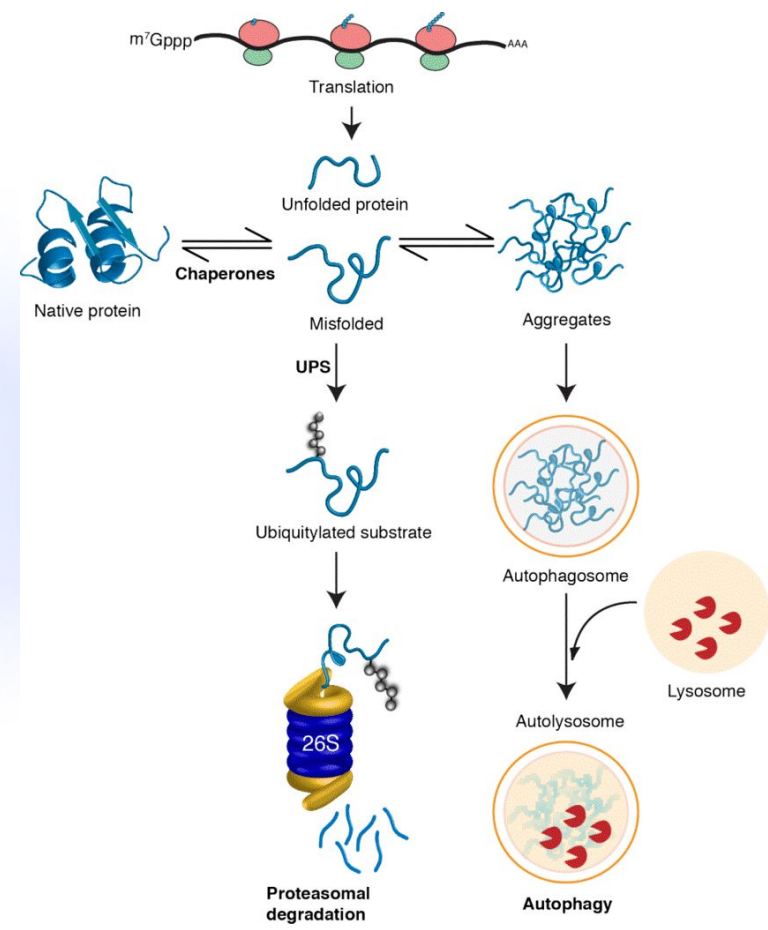
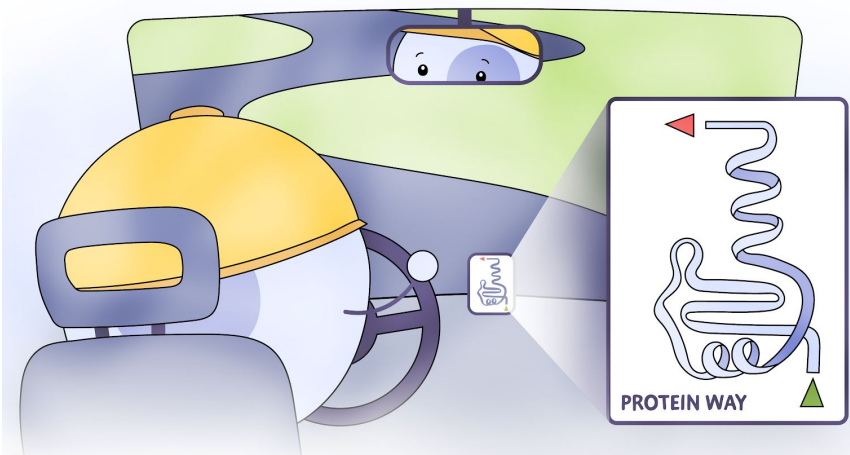
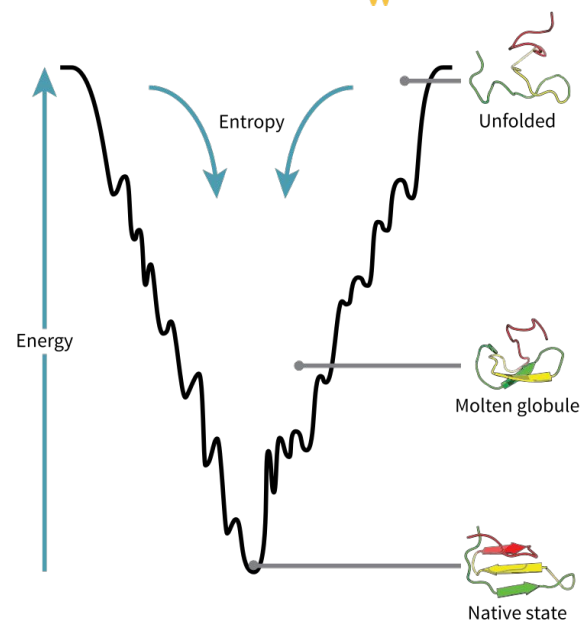
Protein synthesis



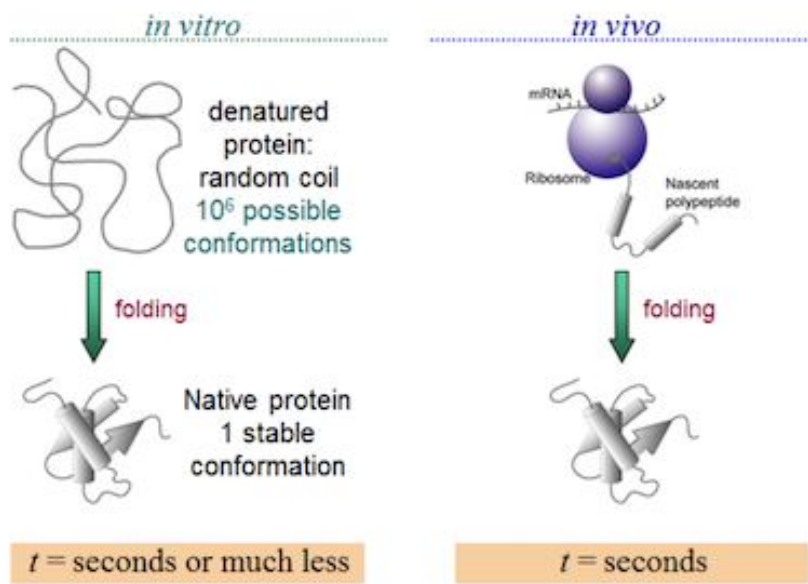
- mRNA carries the genetic code from the DNA in the cell's nucleus to the ribosomes in the cytoplasm.
- Code consists of codons, three-nucleotide sequences that specify which amino acids should be added to the growing protein chain.
- Each codon corresponds to a specific amino acid and the transfer RNA (tRNA) molecules match the codons on the mRNA with the appropriate amino acids.
- As the ribosome moves along the mRNA, it reads the codons and orchestrates the binding of tRNA molecules, which deliver the correct amino acids in the precise sequence dictated by the genetic code.
- This process continues until a stop codon is reached, signaling the end of protein synthesis.



Protein folding



Levinthal paradox



Protein folding: Is it a problem?



What are the factors contributing to folding??

FORCE/BONDS/INTERACTIONS

Levinthal's paradox: "Finding the native folded state of a protein by a random search among all possible configurations can take an enormously long time. Yet proteins can fold in seconds or less."

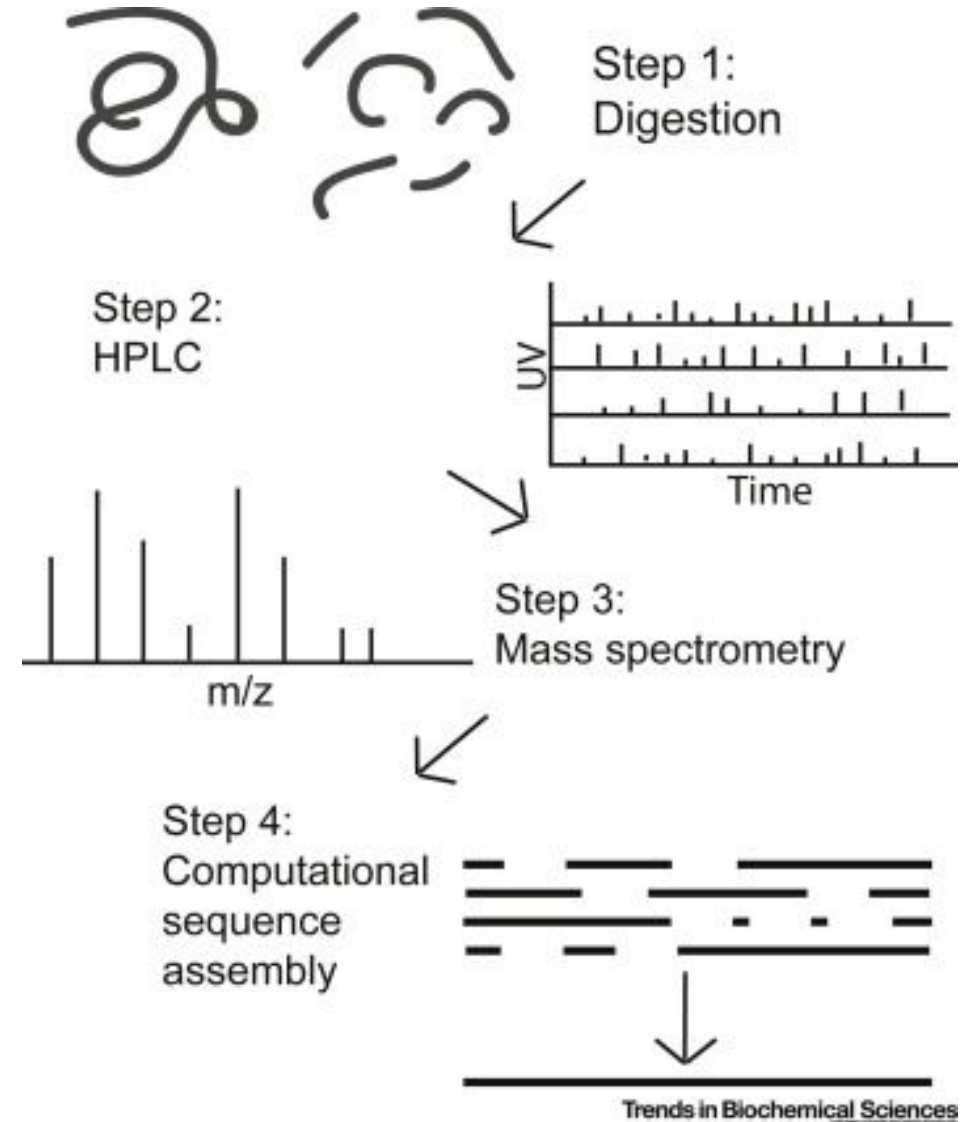
Proc Natl Acad Sci U S A. 1992 Jan 1;89(1):20-2

Anfinsen's "thermodynamic hypothesis": "The native conformation is determined by the totality of interatomic interactions and hence by the amino acid sequence, in a given environment." as stated in 1972 Nobel acceptance speech

Protein sequencing



- Process of determining the precise order of amino acids within a protein molecule.
- **Protein Isolation:** The first step in protein sequencing is to isolate the protein of interest and can be done from a biological sample.
- **Proteolytic Digestion:** To sequence a protein, it must first be broken down into smaller peptides that are easier to analyze.
- **Separation of Peptides:** The resulting mixture of peptides is often complex. High-performance liquid chromatography (HPLC) or other separation techniques are used to separate the peptides based on their properties.
- **Mass Spectrometry (MS):** It identifies the mass of individual peptides and allows for the determination of the amino acid sequence.
- **Tandem Mass Spectrometry (MS/MS):** For more complex proteins, tandem mass spectrometry is used. In this technique, a selected peptide ion is fragmented, and the resulting fragment ions are analyzed.



Can AI help to solve this problem?

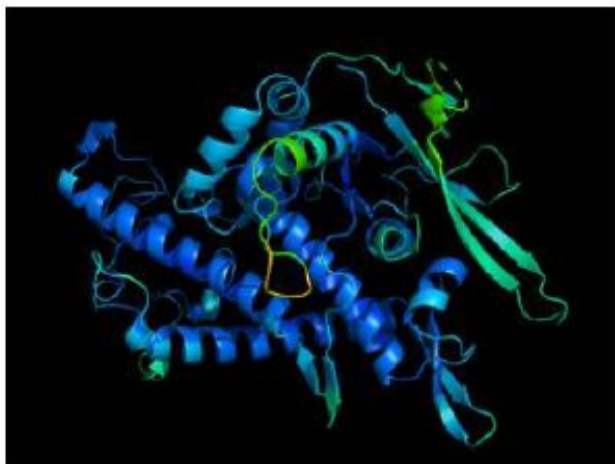
CASP: Critical Assessment of Structure Prediction

NEWS | 30 November 2020

'It will change everything': DeepMind's AI makes gigantic leap in solving protein structures

Google's deep-learning program for determining the 3D shapes of proteins stands to transform biology, say scientists.

Even Calvary



A protein's function is determined by its 3D shape. Credit: DeepMind

An artificial intelligence (AI) network developed by Google AI offshoot DeepMind has made a gargantuan leap in solving one of biology's grandest challenges – determining a protein's 3D shape from its amino acid sequence.

DeepMind's program, called AlphaFold, outperformed around 100 other teams in a biennial protein structure prediction challenge called CASP, short for Critical Assessment of Structure Prediction. The results were announced on 30 November, at the start of the conference – held virtually this year – that takes stock of the exercise.

"This is a big deal," says John Moult, a computational biologist at the University of Maryland in College Park, who co-founded CASP in 1994 to improve computational methods for accurately predicting protein structures. "In some sense the problem is solved."



The ability to accurately predict protein structures from their amino acid sequence would be a huge boon to life sciences and medicine. It would vastly accelerate efforts to understand the building blocks of cells and enable quicker and more advanced

You have full access to this article via Indian
Institute of Technology Jodhpur

Download PDF

Related Articles

[AI protein-folding algorithms solve
structures faster than ever](#)



[The revolution will not be
crystallized: a new method sneaks
through structural biology](#)



[The computational protein designers](#)



[Revolutionary microscopy technique
sees individual atoms for first time](#)



Subjects

[Computational biology and bioinformatics](#)

[Structural biology](#) [Drug discovery](#)

Learn how researchers are
setting the record straight



In partnership with **Potatoes**
UK

Sign up to Nature Briefing

An essential round-up of science news,
opinion and analysis, delivered to your inbox
every weekday.

Email address

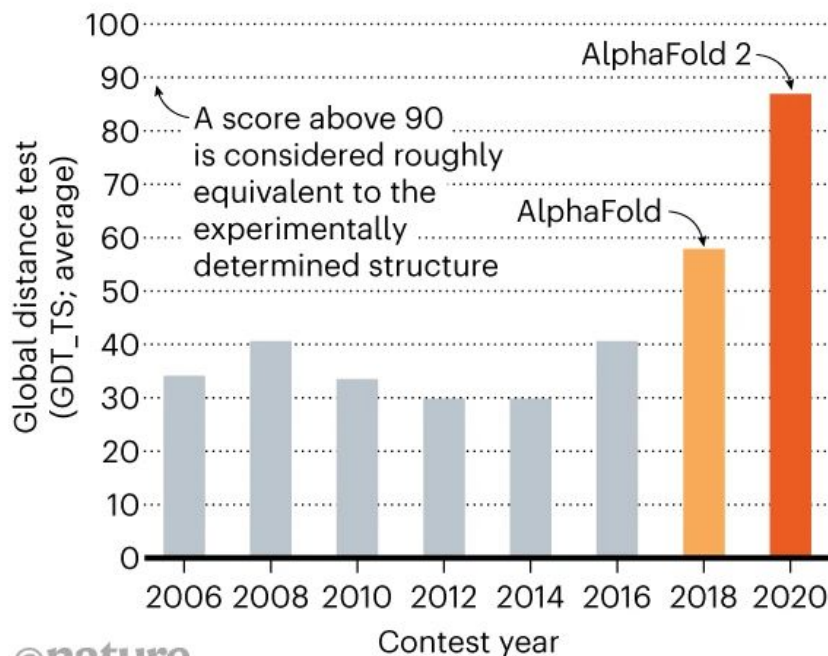
☐ Yes, Sign me up to receive the daily Nature Briefing email. I agree my information will be processed in accordance with the Nature and Springer Nature Limited Privacy Policy.

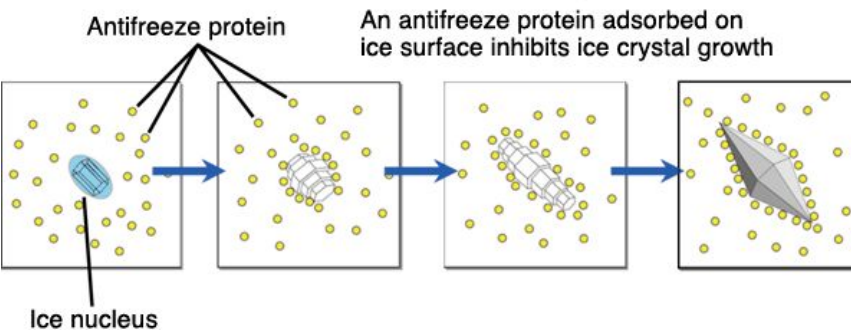
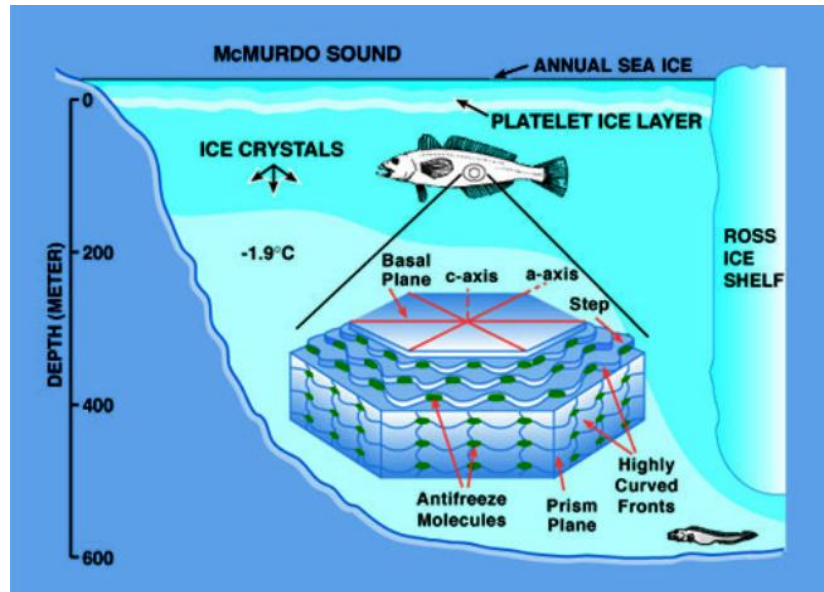
Sign up

"It's a game changer," says Andrei Lupas, an evolutionary biologist at the Max Planck Institute for Developmental Biology in Tübingen, Germany, who assessed the performance of different teams in CASP. AlphaFold has already helped him find the structure of a protein that has vexed his lab for a decade, and he expects it will alter how he works and the questions he tackles. "This will change medicine. It will change research. It will change bioengineering. It will change everything," Lupas adds.

STRUCTURE SOLVER

DeepMind's AlphaFold 2 algorithm significantly outperformed other teams at the CASP14 protein-folding contest – and its previous version's performance at the last CASP.



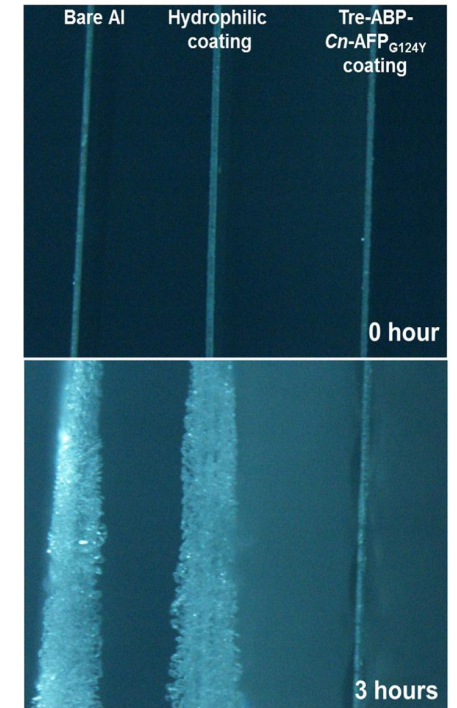
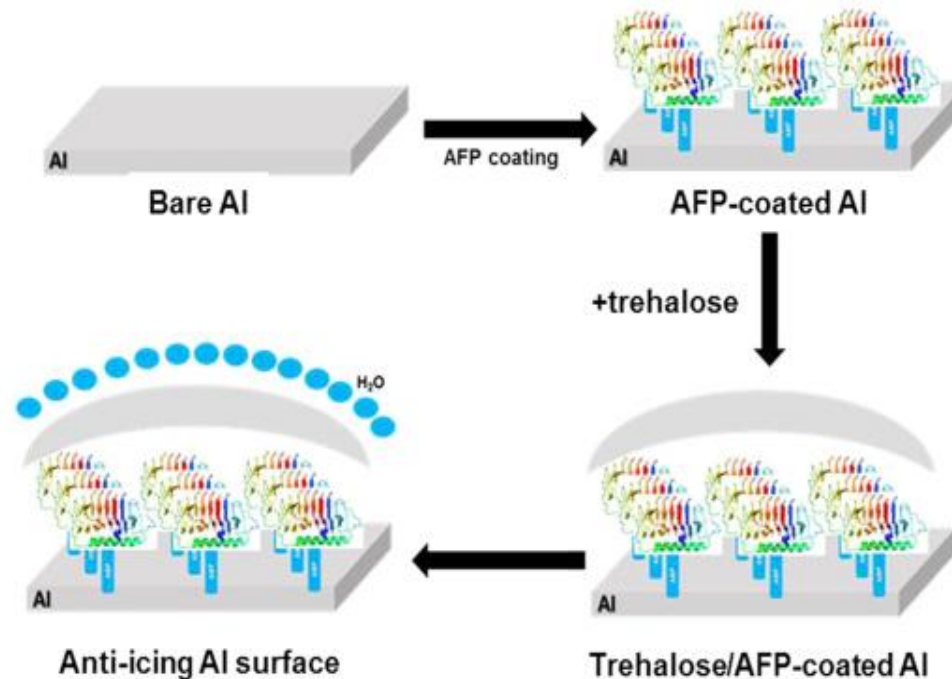


OPEN

Creating Anti-icing Surfaces via the Direct Immobilization of Antifreeze Proteins on Aluminum

Received: 16 January 2015
Accepted: 12 June 2015
Published: 08 July 2015

Yunho Gwak^{1,2}, Ji-in Park^{1,2}, Minjae Kim³, Hong Suk Kim², Myong Jong Kwon², Seung Jin Oh², Young-Pil Kim^{1,3} & EonSeon Jin¹



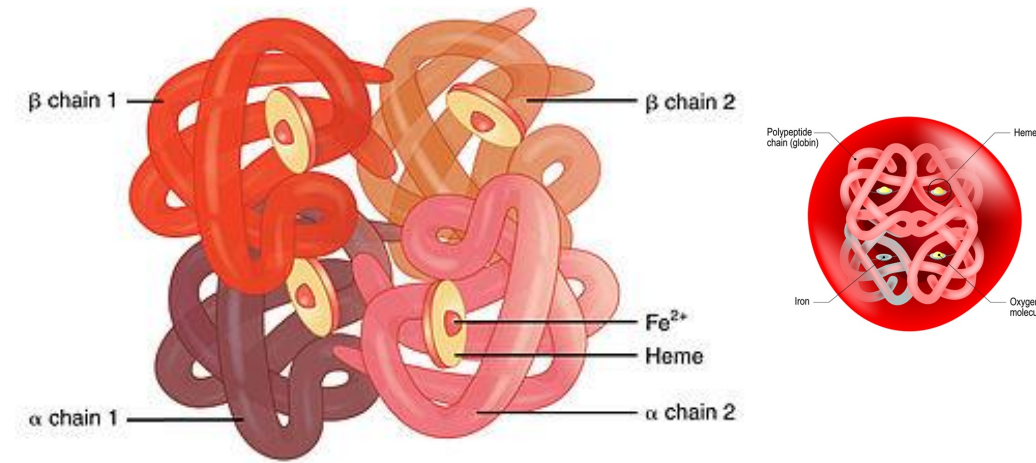
- Anti-freeze proteins, are a class of biomolecules found in various **cold-adapted organisms**, such as **fish, insects, and some plants**, that allow them to survive in freezing environments.
- Some fishes produce AFPs that are **secreted into their blood, body fluids or external surfaces**, allowing them to survive in temperatures below the freezing point of seawater.

Some important proteins



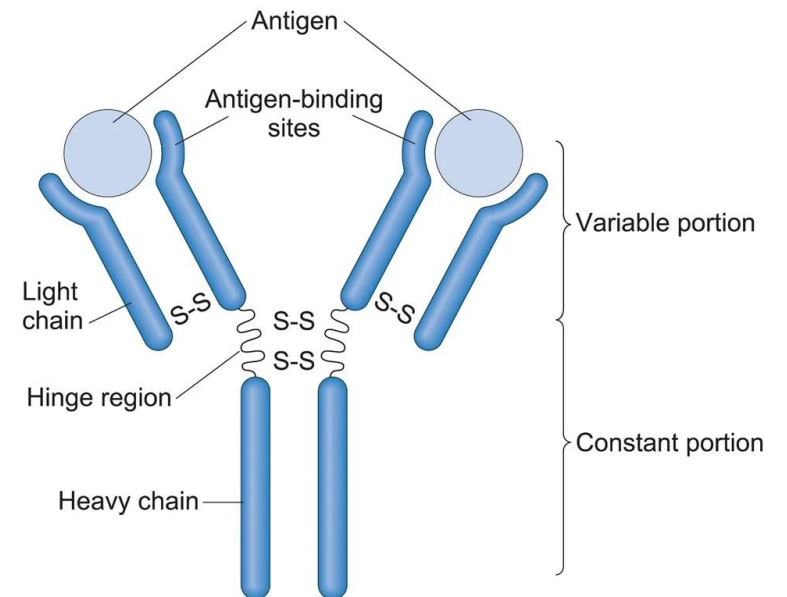
Haemoglobin

- Found in red blood cells that plays a central role in oxygen transport and maintaining the health of the human body.
- Primary function is to bind to oxygen in the lungs and carry it to tissues throughout the body.
- Each hemoglobin molecule can bind to four molecules of oxygen, thanks to its iron-containing heme groups.



Antibodies

- Antibodies are proteins produced by B cells in response to the presence of antigens and they recognize, bind and neutralize these antigens.
- Highly specific and can bind to a specific antigen with a high degree of affinity.
- Antibodies have a Y-shaped structure composed of two heavy chains and two light chains connected by disulfide bonds.

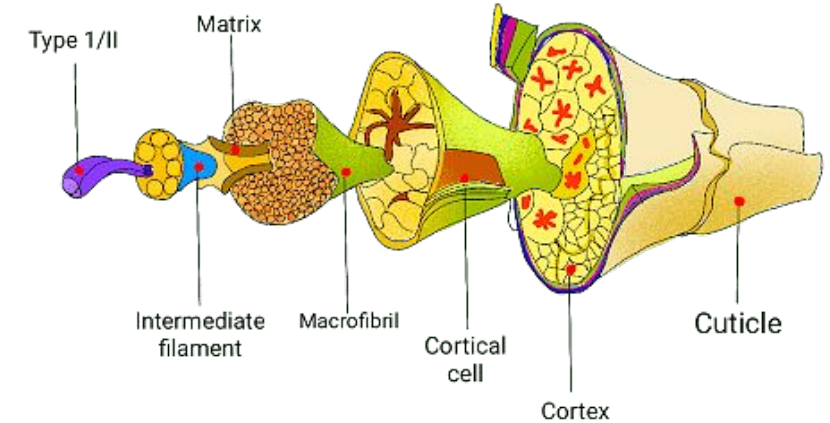


Some important protein



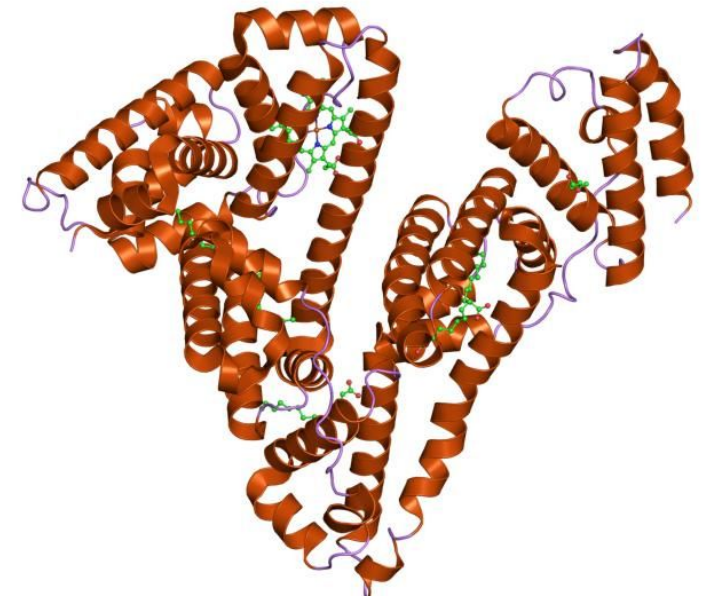
Keratin

- Provides strength, rigidity, and protection to various tissues.
- Keratin is a structural protein that forms the basis of many tough tissues, including skin, hair, nails, feathers, horns, and scales.
- Keratin has a fibrous structure with an elongated, helical shape. These protein chains are rich in the amino acid cysteine, which provide strength and stability.



Albumin

- Albumin is the most abundant protein in human blood plasma synthesized by the liver.
- Maintains blood volume and oncotic pressure, as well as transports various substances.
- Albumin exhibits both alpha-helix and beta-sheet secondary structure which contribute to the protein's overall stability.

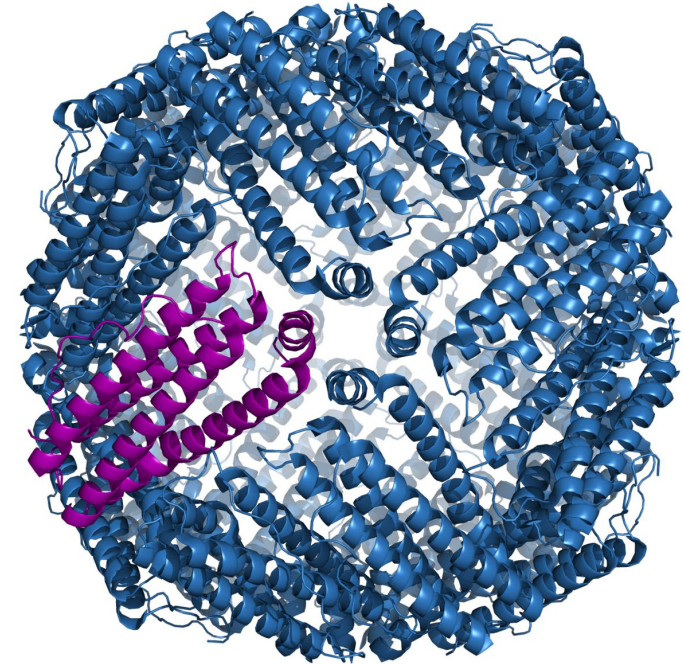


Some important protein



Ferritin

- Ferritin is a universal intracellular protein that stores iron and releases it in a controlled fashion. The protein is produced by almost all living organisms, including archaea, bacteria, algae, higher plants, and animals.
- It is the primary intracellular iron-storage protein in both prokaryotes and eukaryotes, keeping iron in a soluble and non-toxic form. In humans, it acts as a buffer against iron deficiency and iron overload.
- Ferritin is found in most tissues as a cytosolic protein, but small amounts are secreted into the serum where it functions as an iron carrier. Plasma ferritin is also an indirect marker of the total amount of iron stored in the body; hence, serum ferritin is used as a diagnostic test for iron-deficiency anemia. Aggregated ferritin transforms into a toxic form of iron called hemosiderin.
- Ferritin is a globular protein complex consisting of 24 protein subunits forming a hollow nanocage with multiple metal–protein interactions. Ferritin that is not combined with iron is called apoferritin.





Videos to learn more about proteins

<https://www.youtube.com/watch?v=wwTv8TqWC48> What is a protein

<https://www.youtube.com/watch?v=wXvBc-QuCpQ> Lecture by Prof. P. Balaram (IISc Bangalore)